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TITLE :

**ANALYZING STUDENT PERSPECTIVE: A MACHINE LEARNING APPROACH INTO
STUDENTS' ONLINE LEARNING ADAPTABILITY**

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1.0 Acknowledgement

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Furthermore, we are grateful to our parents for their unwavering faith in us, their teachings of never giving up until the task is completed, and their belief in our own abilities and capabilities. Above all, we are grateful to God for being with us and providing us with the knowledge and skills necessary to complete this task. We also appreciate all of our friends for their moral support.

2.0 Abstract

This study explores the constantly evolving field of online learning, using a strong classification-based technique to analyze students' perspectives of adaptability in online education. To truly evaluate the efficiency of online learning, we meticulously collected data about students' adaptability level on Kaggle and thoroughly analyzed it using five different machine learning models: Fast Tree, L1 Regularization, XGBoost, CatBoost, and LightGBM. Interestingly, among used models, the Fast Tree model achieved the best accuracy of 0.91, outperforming other models. The results reveal important patterns and trends and offer detailed insights into the complex elements impacting students' success in online learning. This study offers useful applications for improving online learning strategies and provides actionable knowledge that could improve the online learning adaptability among students. The results also highlight the necessity of focused interventions and instructional strategies, recognising the multitude of variables influencing students' ability to adapt. These insights significantly contribute to continuing discussions on improving student outcomes in the online environment and are a useful tool for educators and institutions navigating the ever-changing world of digital education. This study establishes the foundation for further research as online education develops, promoting continuous improvement of approaches and comprehension in this quickly evolving field. For educators and policymakers looking to maximize the benefits of online learning and create effective learning environments in the digital age, this study is invaluable.

Keywords: sklearn classification, probability distributions, online learning, statistical analysis, machine learning, xgboost, catboost, fast tree, L1 regularization, lightgbm.

3.0 Introduction

Embarking on the intersection of Probability and Statistics with the dynamic realm of online education, our study, "Analyzing Student Perspective: A Machine Learning Approach into Students' Online Learning Adaptability," delves into the intricate nuances of students' capacity to navigate online learning environments. Much like statistical graphs and probability distributions prove instrumental in understanding and predicting heart diseases, our research seeks to unravel patterns and trends shaping students' success in online education. By conducting statistical analyses, we aim to identify key factors influencing adaptability, providing educators and institutions with crucial insights to enhance teaching methodologies and foster supportive online learning ecosystems.

This study uses the predictive power of five different machine learning models, XGBoost, CatBoost, FastTree, L1 Regularization, and LightGBM to understand students' adaptation to online learning. The highly accurate XGBoost algorithm uses statistical graphs and probability distributions to reveal intricate links in the data. It is particularly good at identifying nonlinear dependencies. CatBoost, designed for categorical features, uses comparable statistical techniques to handle a wide range of student characteristics with ease. FastTree, known for its effectiveness, speeds up model training and assessment, making it easier to investigate scenarios pertaining to flexibility. L1 Regularization adds sparsity to feature selection, highlighting important features that facilitate an understandable analysis. Histogram-based learning is used by LightGBM, which is renowned for its scalability and efficiency, guaranteeing quick and economical analysis.

After collecting data from Kaggle about Students' Adaptability Level in Online Education, it is found out that only 8% of 1205 students are able to highly adapt to online learning, about 52% moderately adapt and 40% hardly adapt to online learning. There are many opportunities in online education, but they must be utilized properly.

As a result, the major contributions suggested in this study are:

- Sociodemographic variables may be very important in ensuring that online learning runs smoothly. Thus, estimating the level of adaptability of the student based on these variables aids in understanding the policy makers and enables them to take the required actions to lessen the problems.

- A comparison of the model's performance is done in order to comprehend the models' predictive capability.
- To predict student adaptability, this study employed XGBoost, CatBoost, FastTree, L1 Regularization, and LightGBM. The Fast Tree model produced the best prediction accuracy of 0.91, outperforming other models.

4.0 Dataset Acquisition

- **Dataset Description**

The dataset of this study is collected from Kaggle, titled Student Adaptability Level on Online Education that can be collected from the following link: <https://www.kaggle.com/datasets/mdmahmudulhasansuzan/students-adaptability-level-in-online-education?resource=download>

The dataset has been inserted as headers for data to store for each row. In this study, the dataset contained 14 attributes which are gender, age, education level, institution level, institution type, IT student, location, load-shedding, financial condition, internet type, network type, class duration, self LMS, device and adaptivity level. The dataset has no missing values and it is ensured complete.

- **Exploratory Data Analysis (EDA)**

EDA is an approach of analyzing data sets to highlight their key features, frequently using techniques that are visually appealing. Finding patterns in the data, recognising outliers, gaining insights into its underlying structure, and developing hypotheses for more research are the goals of Exploratory Data Analysis (EDA).

Firstly, the dataset is loaded by using **pandas** to read the dataset into a DataFrame.

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
```

```
#Import the dataset file
data=pd.read_csv("students_adaptability_level_online_education.csv")
data
```

	Gender	Age	Education Level	Institution Type	IT Student	Location	Load-shedding	Financial Condition	Internet Type	Network Type	Class Duration	Self Lms	Device	Adaptivity Level
0	Boy	21-25	University	Non Government	No	Yes	Low	Mid	Wifi	4G	3-6	No	Tab	Moderate
1	Girl	21-25	University	Non Government	No	Yes	High	Mid	Mobile Data	4G	1-3	Yes	Mobile	Moderate
2	Girl	16-20	College	Government	No	Yes	Low	Mid	Wifi	4G	1-3	No	Mobile	Moderate
3	Girl	11-15	School	Non Government	No	Yes	Low	Mid	Mobile Data	4G	1-3	No	Mobile	Moderate
4	Girl	16-20	School	Non Government	No	Yes	Low	Poor	Mobile Data	3G	0	No	Mobile	Low

Secondly, the dataset is inspected by checking the first few rows of the dataset using the **head()** method to understand its structure and contents.

```
df.head()
```

	Gender	Age	Education Level	Institution Type	IT Student	Location	Load-shedding	Financial Condition	Internet Type	Network Type	Class Duration	Self Lms	Device	Adaptivity Level
0	Boy	21-25	University	Non Government	No	Yes	Low	Mid	Wifi	4G	3-6	No	Tab	Moderate
1	Girl	21-25	University	Non Government	No	Yes	High	Mid	Mobile Data	4G	1-3	Yes	Mobile	Moderate
2	Girl	16-20	College	Government	No	Yes	Low	Mid	Wifi	4G	1-3	No	Mobile	Moderate
3	Girl	11-15	School	Non Government	No	Yes	Low	Mid	Mobile Data	4G	1-3	No	Mobile	Moderate
4	Girl	16-20	School	Non Government	No	Yes	Low	Poor	Mobile Data	3G	0	No	Mobile	Low

Thirdly, summary statistics of the dataset is generated by using descriptive statistics functions like **describe()**.

```
data.describe()
```

	Gender	Age	Education Level	Institution Type	IT Student	Location	Load-shedding	Financial Condition	Internet Type	Network Type	Class Duration	Self Lms	Device	Adaptivity Level
count	1205	1205	1205	1205	1205	1205	1205	1205	1205	1205	1205	1205	1205	1205
unique	2	6	3	2	2	2	2	3	2	3	3	2	3	3
top	Boy	21-25	School	Non Government	No	Yes	Low	Mid	Mobile Data	4G	1-3	No	Mobile	Moderate
freq	663	374	530	823	901	935	1004	878	695	775	840	995	1013	625

```
# numerical and categorical columns
numeric_features = [feature for feature in data.columns if data[feature].dtype != 'O']
categorical_features = [feature for feature in data.columns if data[feature].dtype == 'O']

# print columns
print('We have {} numerical features : {}'.format(len(numeric_features), numeric_features))
print('\nWe have {} categorical features : {}'.format(len(categorical_features), categorical_features))
```

```
We have 0 numerical features : []
```

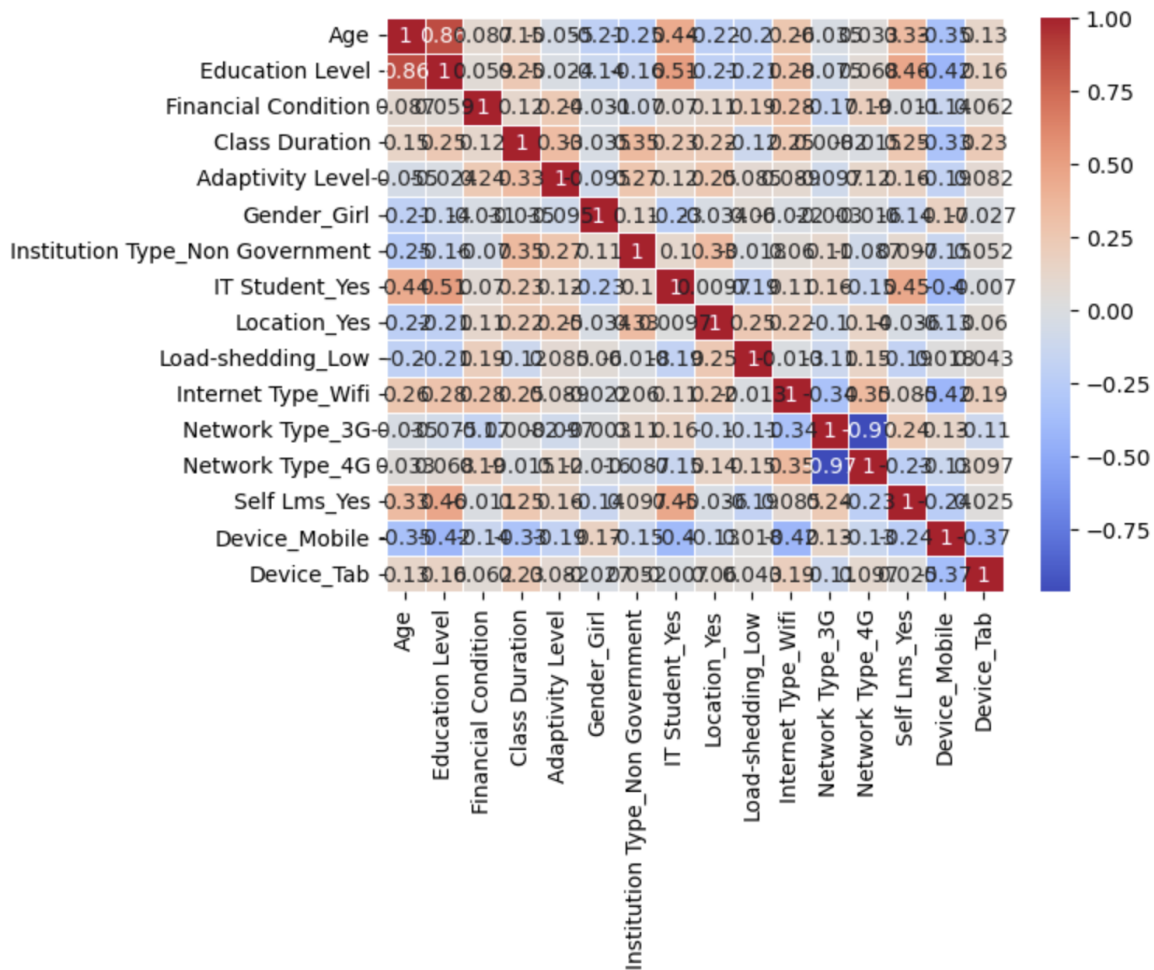
```
We have 14 categorical features : ['Gender', 'Age', 'Education Level', 'Institution Type', 'IT Student', 'Location', 'Load-shedding', 'Financial Condition', 'Internet Type', 'Network Type', 'Class Duration', 'Self Lms', 'Device', 'Adaptivity Level']
```

As a result, the summary statistics show count, unique, top and freq of every feature in the dataset. Count represents the number of non-null values in the column. It gives an indication of how many values are present and whether there are missing values in the dataset. Unique represents the number of unique values in the column. It gives insight into the diversity or variability of values in the column. Top represents the most frequently occurring (mode) value in the column. It provides information about the dominant category or value in the dataset. Freq represents the frequency of occurrence of the top value. It indicates how many times the top value appears in the dataset.

Next, missing values are handled by using methods like **isna()**.

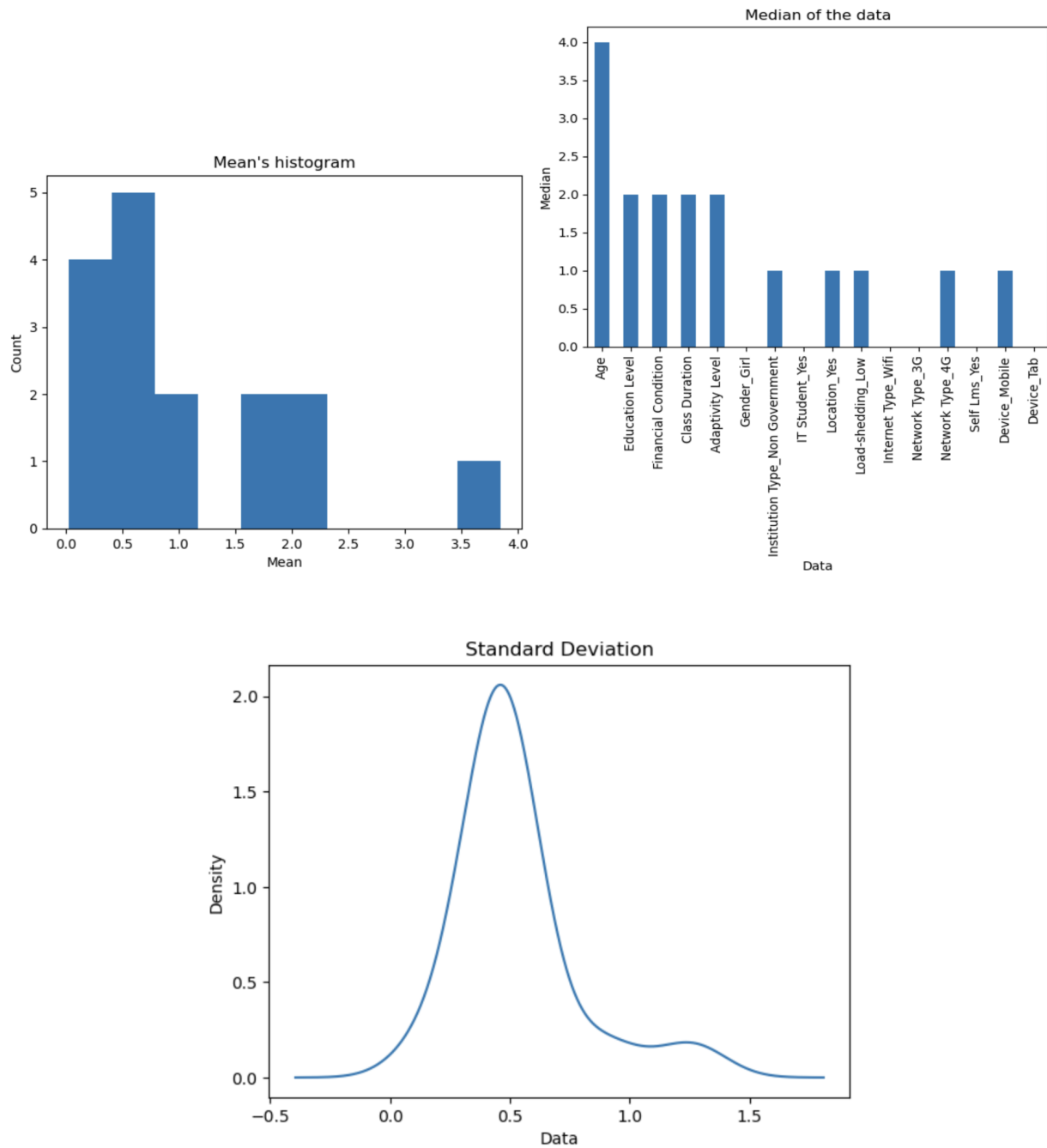
```
data.isna().sum()
Gender      0
Age         0
Education Level  0
Institution Type  0
IT Student  0
Location    0
Load-shedding  0
Financial Condition  0
Internet Type  0
Network Type  0
Class Duration  0
Self Lms    0
Device      0
Adaptivity Level  0
dtype: int64
```

Lastly, correlation analysis is conducted by examining the correlations between features using functions like **corr()** and visualizing them using heatmaps.



Correlation Matrix

Basic statistics like mean, median and standard deviation are visualized using graphs to understand more about the distribution of the dataset.



5.0 Dataset Preparation

- **Dataset Distribution**

In the exploration of the machine learning models, XGBoost, CatBoost, FastTree, L1 Regularization, and LightGBM, the distribution of the dataset into training and testing sets is a critical step for achieving accurate and reliable results.

For this analysis, data are split to ratio of 80%-20%, with 80% of the data earmarked for training and the remaining 20% for testing. This division strikes a balance between providing each model with sufficient data to learn intricate patterns and preserving a separate subset for unbiased evaluation. The split ratio aligns with this study analysis goals, emphasizing model performance in real-world scenarios.

```
#Split the data
X=dataframe.drop(['Adaptivity Level'], axis=1)
y=dataframe['Adaptivity Level']

X_test, X_train, y_test, y_train=train_test_split(X, y, test_size=0.2, random_state=42)

#Display the shape of each step
print("X_test shape: ", X_test.shape)
print("X_train shape: ", X_train.shape)
print("y_test shape: ", y_test.shape)
print("y_train shape: ", y_train.shape)
```

Figure 3: Test and train of dataset where random_state is utilized

To ensure the reproducibility of the results, a random_state parameter is employed. This parameter allows a specific random seed to be set, ensuring that the random distribution of data remains consistent across different runs.

- **Data Pre-processing**

Identification of unique values are made to understand the range of values present in each feature and identify potential issues, such as missing or unexpected values.

```
for i in dataframe.columns:
    print(i, ":", dataframe[i].unique().tolist())
```

Figure 4: shows code for identify unique values

The dataset from Kaggle was encoded to convert categorical data into numerical format, making it suitable for machine learning models that require numerical input.

```
from sklearn.preprocessing import LabelEncoder

categorical_columns=['Gender', 'Age', 'Education Level', 'Institution Type', 'IT Student', 'Location', 'Load-shedding', 'Financial Condition',

#Create LabelEncoder instances
label_encoder=LabelEncoder()

#Iterate through each column and apply LabelEncoder
for col in categorical_columns:
    if col in dataframe.columns:
        dataframe[col] = label_encoder.fit_transform(dataframe[col])

#Print
dataframe.head()
```

Figure 5: shows encoding of dataset

From the data pre-processing, the study found no missing values of the dataset, thus no potential issues regarding the data and the data is ensured complete. The dataset is prepared in a way that maximizes the model's ability to learn patterns, minimizes biases, and ensures reproducibility, which aligns with best practices in machine learning.

● Experiment Repetition

```
# Iterate over random_state values and perform experiments
for random_state_val in range(1, 10001):
    # Split the data
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=random_state_val)

    # Fit the model
    model.fit(X_train, y_train)

    # Predict the labels
    y_pred = model.predict(X_test)

    # Append classification report to the list
    report = classification_report(y_test, y_pred, output_dict=True)
    report['Random State'] = random_state_val
    classification_reports.append(report)
```

Figure 6: Shows iteration of dataset from random_state=1 to random_state=10000

This repetition results in 10000 classification reports of the dataset that give precision, f1-score and accuracies of each iteration. This repetition helps in minimizing the impact of randomness and ensures that the model's performance is not overly dependent on a specific data split.

After processing the dataset, the study gains details of all 14 attributes of the dataset. Table 1 shows all the attributes used in this study.

Table 1: Details of all 14 attributes of the dataset

	count	unique	top	freq	0
Gender	1205	2	Boy	663	Boy, Girl
Age	1205	6	21-25	374	21-25, 16-20, 11-15, 26-30, 6-10, 1-5
Education Level	1205	3	School	530	University, College, School
Institution Type	1205	2	Non Government	823	Non Government, Government
IT Student	1205	2	No	901	No, Yes
Location	1205	2	Yes	935	Yes, No
Load-shedding	1205	2	Low	1004	Low, High
Financial Condition	1205	3	Mid	878	Mid, Poor, Rich
Internet Type	1205	2	Mobile Data	695	Wifi, Mobile Data
Network Type	1205	3	4G	775	4G, 3G, 2G
Class Duration	1205	3	1-3	840	3-6, 1-3, 0
Self Lms	1205	2	No	995	No, Yes
Device	1205	3	Mobile	1013	Tab, Mobile, Computer
Adaptivity Level	1205	3	Moderate	625	Moderate, Low, High

6.0 Methodology

- XGBoost

The Gradient Boosting algorithm is an ensemble learning technique that is applied with XGBoost (Extreme Gradient Boosting). For many machine learning issues, XGBoost is a well-liked solution, particularly for structured data categorization and regression. In contrast to the Gradient Boosting approach, XGBoost minimizes the model's overall error by employing several trees. In order to accelerate model training, it also enables the usage of additional processors. These characteristics enable great accuracy, speed, and scalability with XGBoost.

- L1 Regularization

L1 regularization is a technique in machine learning employed to prevent overfitting by adding a penalty term to the standard linear regression cost function. The penalty term is based on the absolute values of the model coefficients, encouraging sparsity and driving some coefficients to precisely zero. This sparsity promotes feature selection, simplifying the model and enhancing interpretability. L1 regularization is particularly valuable when dealing with high-dimensional datasets, as it aids in selecting the most relevant features and preventing overfitting by simplifying the model.

- CatBoost

CatBoost is an outstanding performance open-source library for gradient boosting on decision trees that can be applied to classification, regression, and ranking tasks. CatBoost achieves excellent performance on huge and complicated sets of data with categorical features by combining ordered boosting, random permutations, and gradient-based optimization techniques.

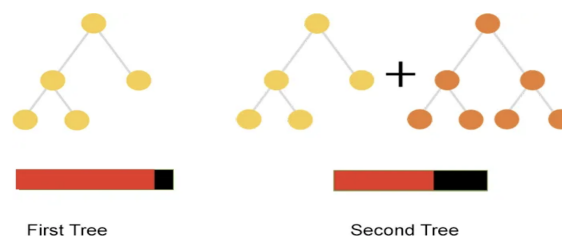


Figure 7: Shows how CatBoost works

- LightGBM

LightGBM was created to outperform XGBoost in terms of performance and training time. It does not grow level-wise; rather, it grows leaf-wise. A leaf-wise development technique entails developing the tree by setting a minimum number of observations or a maximum depth for each of the tree's leaf nodes. With this approach, the tree is grown until it reaches the predetermined depth or number of leaf nodes. The tree's depth and number of leaf nodes are predefined. Compared to the level-wise growth approach, this strategy is more effective and uses less memory and takes less time to create the tree.

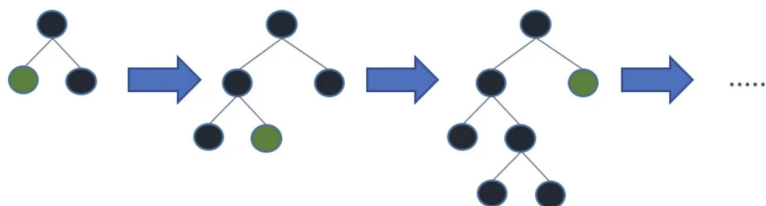


Figure 8: Shows Leaf-wise Tree Growth

- Fast Tree

FastTree is a machine learning algorithm designed for rapid decision tree construction, emphasizing efficiency and speed. It belongs to the family of ensemble methods, specifically random forests, where multiple decision trees are trained independently and their outputs are combined to enhance predictive accuracy. FastTree introduces two key innovations to expedite the training process: a top-down tree-building strategy that efficiently identifies informative split points, and a randomized selection of feature subsets for each tree to encourage diversity and prevent overfitting. By optimizing the tree-growing process, FastTree achieves faster training times while maintaining competitive predictive performance.

Performance Metrics Selection: Classification Metrics

When it comes to choosing which metrics for the machine learning model, there are a few different metrics for tasks like classification and regression. There are also different metrics that are useful for multiple tasks like precision recall. While using different metrics for machine learning performance evaluation, this study is able to improve the performance of the models. Without a proper evaluation only depending on the accuracy may bring problems to machine learning results

such as poor results. Due to that, in order to avoid such problems, Classification Metrics are used which consist of Accuracy, Precision, F1 scores, and Sensitivity.

Machine learning models' performance is assessed using classification metrics such as Sensitivity, F1 Score, Accuracy, and Precision, particularly in classification tasks. With accuracy as the ratio of properly predicted instances to the total, the model's correctness is generally gauged. Precision highlights the applicability of the model's predictions by measuring how well it can detect positive cases. Precision and Recall (Sensitivity) are combined into one metric called F1 Score, which provides a fair evaluation of a model's total performance. Sensitivity quantifies how well the model captures every good example. Together, these metrics provide a thorough understanding of a model's advantages and disadvantages, which helps optimize algorithms for particular use cases—in this case, determining the level of adaptability of students in online learning—by illuminating the trade-offs between false positives and false negatives.

Statistical Analysis

Statistical analysis is a method used to collect, analyze, interpret, and present data. It involves the use of statistical techniques to uncover patterns, trends, relationships, and insights within datasets. Several important statistical techniques are used to analyze the findings of tests carried out on five different machine learning models—XGBoost, CatBoost, FastTree, L1 Regularization, and LightGBM—across a wide range of random states ranging from 1 to 10000. The computation of mean, minimum, maximum, and mode values enables the identification of prevailing patterns and key tendencies. While confidence intervals provide information on the dependability of performance measures, standard deviation is calculated to assess the consistency of model performance. The goal of this thorough statistical study is to reveal the machine learning models' complex behaviors and varying relative performances across a range of data distributions and experimental setups.

- Mean

The mean, also known as the average, is a measure of central tendency. It is calculated by summing up all the values involved and dividing by the total number of values.

$$\underline{X} = \frac{\sum x}{N}$$

$\underline{X}$	=	Value of Mean
Σx	=	Total sum of the values involved
N	=	Total number of values

- Standard Deviation

Standard deviation is a measure of the dispersion or spread of a set of values. It quantifies how much individual data points deviate from the mean (average) of the dataset. In other words, it indicates the average distance of each data point from the mean.

$$\text{Standard deviation} = \sqrt{\frac{\sum_{i=1}^n (x_i - \underline{X})^2}{n}}$$

x_i	=	Value of each individual data point
$\underline{X}$	=	Value of Mean(average)
n	=	Total number of data points

- Confidence Interval

A confidence interval is a range of values derived from a sample statistic that is likely to contain the true population parameter with a certain level of confidence. It provides a measure of the uncertainty or variability associated with estimating population parameters based on sample data.

$$\underline{X} - Z_{\frac{\alpha}{s}} \left(\frac{\text{Standard Deviation}}{\sqrt{N}} \right) < Z < \underline{X} + Z_{\frac{\alpha}{s}} \left(\frac{\text{Standard Deviation}}{\sqrt{N}} \right)$$

$\underline{X}$	=	Value of Mean
$Z_{\frac{\alpha}{s}}$	=	Critical value of Z
N	=	Total number of numbers

- Min, Max, Mode

Minimum (min), Maximum (max), and Mode offer further information about the distribution and variation of machine learning models' accuracy.

1. Minimum (Min):

- The lowest performance attained by any model in the set is indicated by the minimal accuracy.
- It assists in identifying the least effective model, which would need more research or development.
- Knowing the lowest accuracy provides information about the models' worst-case performance.

2. Maximum (Max):

- The top performance attained by any model in the set is indicated by the maximum accuracy.
- It draws attention to the top-performing model and its value as a comparative standard.
- The top bound of performance that can be achieved with the provided dataset and modeling methodologies by identifying the maximum accuracy.

3. Mode:

- Among the models, the accuracy value that occurs the most frequently is represented by the mode.
- It assists in determining the average performance level that several models share.
- Analysing the mode helps identify trends or groups of models with comparable performance.
- When there are several modes, it means that there are several model groups with varying degrees of performance.

7.0 Results

- XGBoost

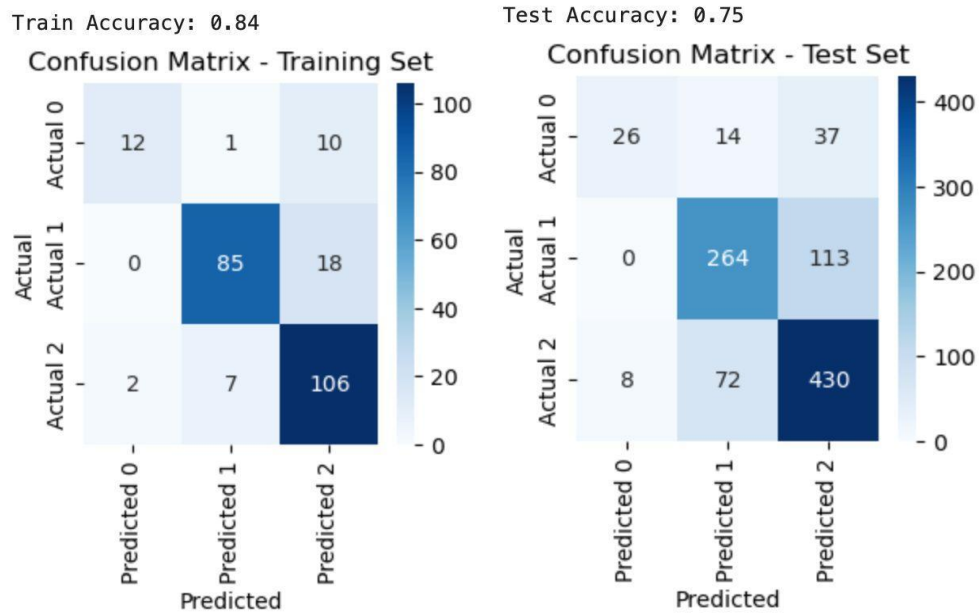


Figure 9: Model XGBoost Training Set and Test Set Distribution on Student Online Learning Adaptability

Table 2 : Classification Report for XGBoost Model on Student Online Learning Adaptability

Classification Report:					
	precision	recall	f1-score	support	
High	0.76	0.34	0.47	77	
Low	0.75	0.70	0.73	377	
Moderate	0.74	0.84	0.79	510	
accuracy			0.75	964	
macro avg	0.75	0.63	0.66	964	
weighted avg	0.75	0.75	0.74	964	

- LightGBM

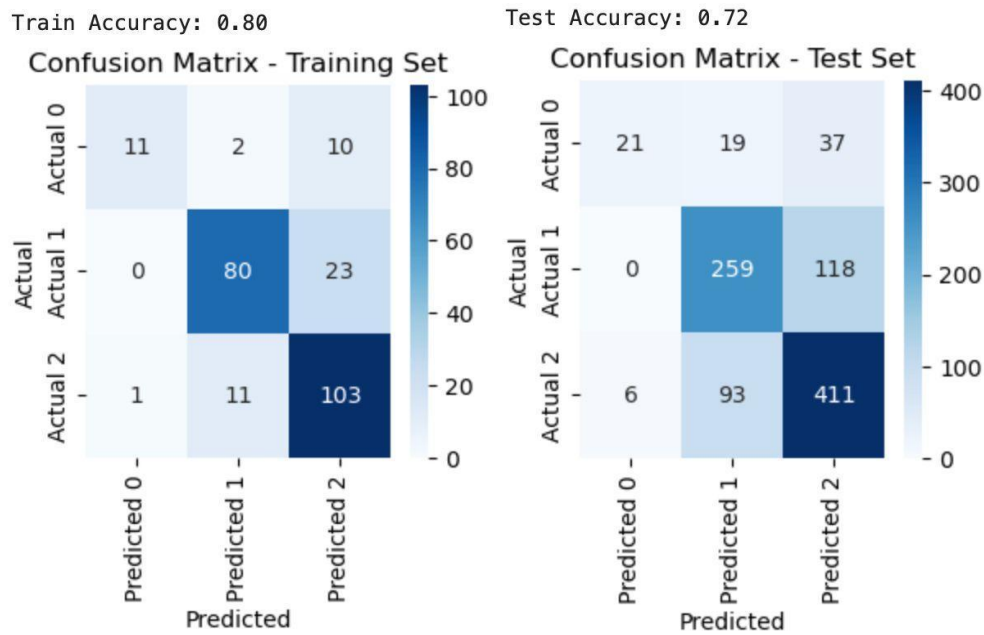


Figure 10: Model LightGBM Training Set and Test Set Distribution on Student Online Learning Adaptability

Table 3: Classification Report for LightGBM Model on Student Online Learning Adaptability

Classification Report:

	precision	recall	f1-score	support
High	0.78	0.27	0.40	77
Low	0.70	0.69	0.69	377
Moderate	0.73	0.81	0.76	510
accuracy			0.72	964
macro avg	0.73	0.59	0.62	964
weighted avg	0.72	0.72	0.71	964

- CatBoost

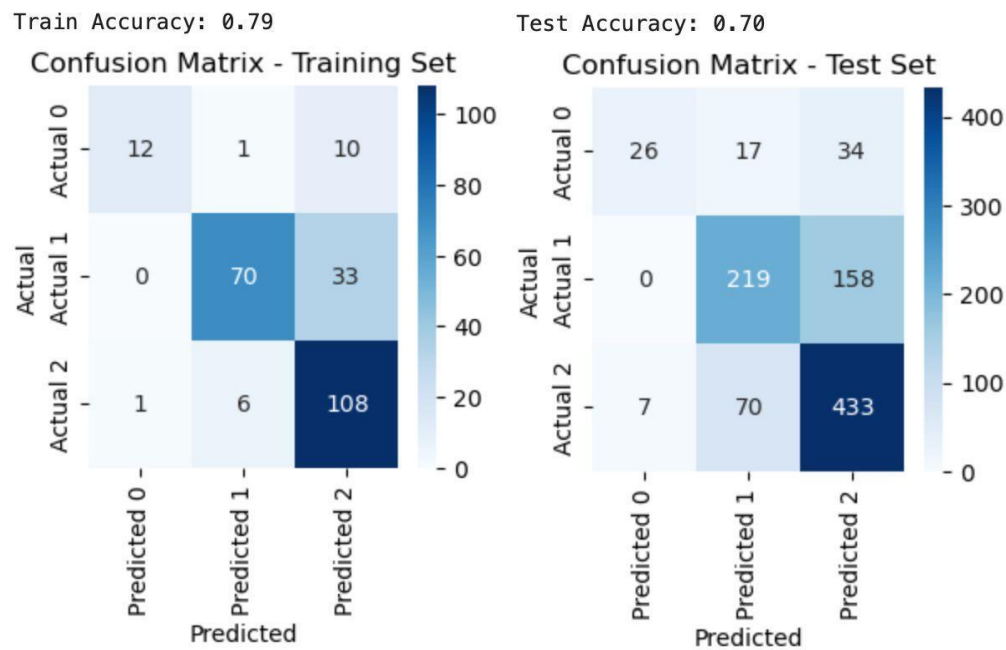


Figure 11: Model CatBoost Training Set and Test set Distribution on Student Online Learning Adaptability

Table 4: Classification Report for CatBoost Model on Student Online Learning Adaptability

Classification Report:

	precision	recall	f1-score	support
High	0.79	0.34	0.47	77
Low	0.72	0.58	0.64	377
Moderate	0.69	0.85	0.76	510
accuracy			0.70	964
macro avg	0.73	0.59	0.63	964
weighted avg	0.71	0.70	0.69	964

- L1 Regularization

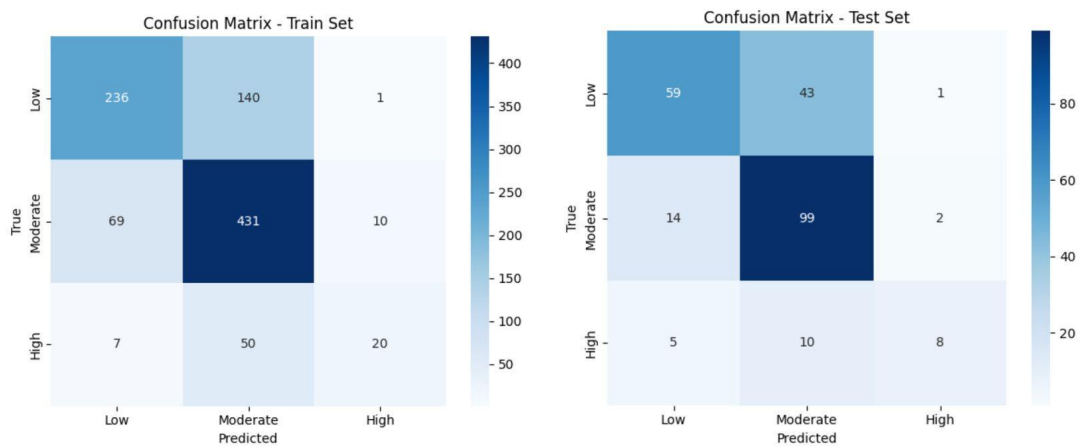


Figure 12: Model L1 Regularization Train Set and Test Set Distribution on Student Online Learning Adaptability

Table 5: Classification Report for L1 Regularization Model on Student Online Learning Adaptability

Accuracy: 0.6887966804979253

Classification Report:

	precision	recall	f1-score	support
1	0.76	0.57	0.65	103
2	0.65	0.86	0.74	115
3	0.73	0.35	0.47	23
accuracy			0.69	241
macro avg	0.71	0.59	0.62	241
weighted avg	0.70	0.69	0.68	241

- FastTree

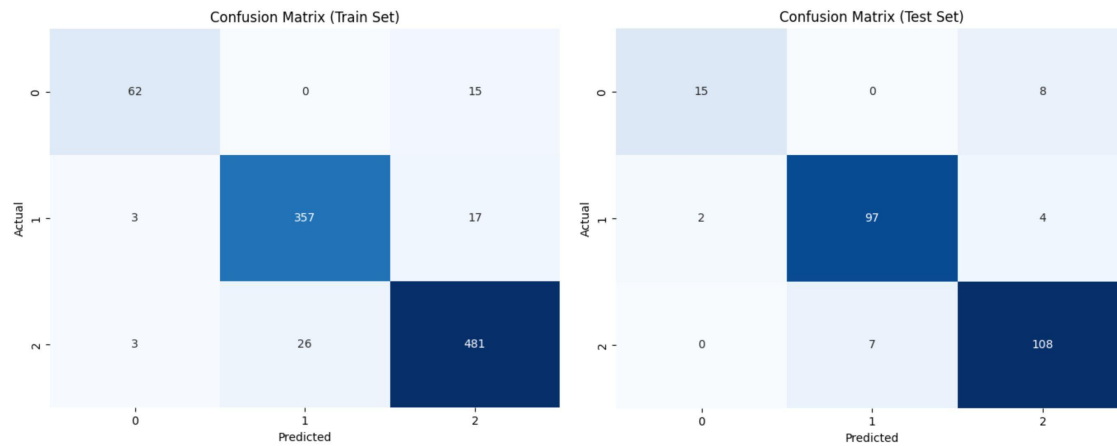


Figure 13: Model FastTree Train Set and Test Set Distribution on Student Online Learning Adaptability

Table 6: Classification Report for FastTree Model on Student Online Learning Adaptability

Accuracy: 0.9128630705394191

Classification Report:

	precision	recall	f1-score	support
High	0.88	0.65	0.75	23
Low	0.93	0.94	0.94	103
Moderate	0.90	0.94	0.92	115
accuracy			0.91	241
macro avg	0.91	0.84	0.87	241
weighted avg	0.91	0.91	0.91	241

Table 7: Summary of Model Performance Metrics

Metric	Value
Mean Accuracy	0.75
Minimum Accuracy	0.69
Maximum Accuracy	0.91
Mode Accuracy	0.69
Standard Deviation	0.08424037428963
95% Confidence Interval	(0.6466548720825511, 0.8558513519838387)

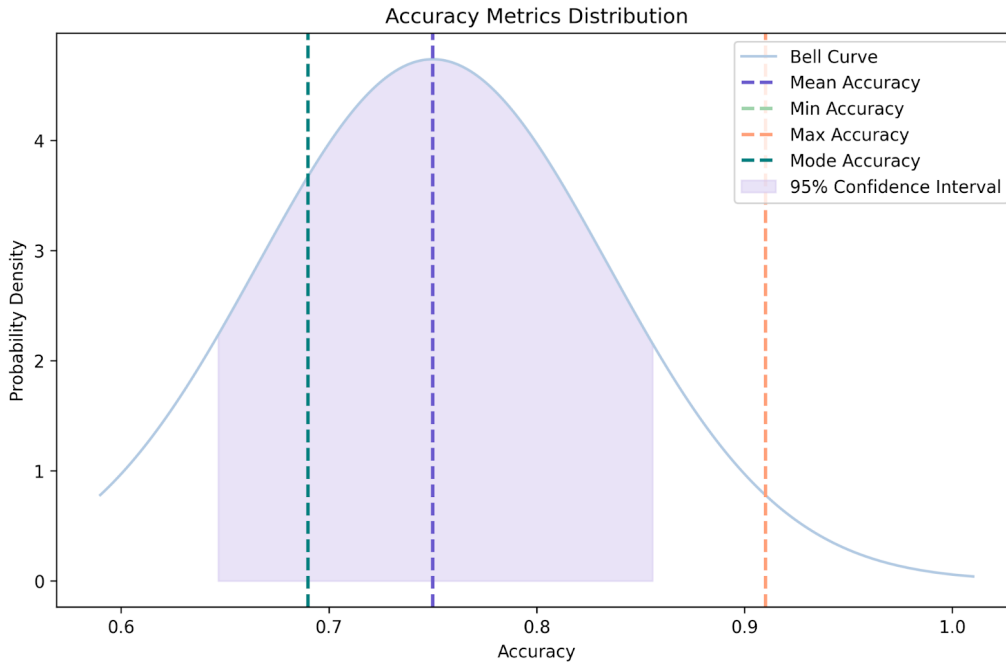


Figure 14: Distribution of Model accuracy metrics

Table 8: T-test for comparing the means of accuracy of each Model

Model 1	Model 2	T-statistics	P-value
L1 Regularization	XGBoost	-2.7020	0.0000
L1 Regularization	CatBoost	-40279	0.0000
L1 Regularization	LightGBM	-1.2084	0.0000
L1 Regularization	FastTree	-9.9074	0.0000
XGBoost	CatBoost	4.5034	0.0000
XGBoost	LightGBM	2.7020	0.0000
XGBoost	FastTree	-inf	0.0000
CatBoost	LightGBM	-1.2737	0.0000
CatBoost	FastTree	-1.8914	0.0000
LightGBM	FastTree	-1.7113	0.0000

8.0 Discussion of Results

The effectiveness of machine learning models can be evaluated using a variety of metrics. The most crucial factors in evaluating a model's performance are its accuracy, precision, recall, and F1 score. The precision, recall, F1-Score, and accuracy scores are determined using the confusion matrix value that is produced during the model's testing. The formulas are the one used for the computation of the confusion matrix for each model.

$$\begin{aligned} Precision &= \frac{TP}{TP + FP} \\ Recall &= \frac{TP}{TP + FN} \\ Accuracy &= \frac{TP + TN}{TP + FP + FN + TN} \\ F1 - Score &= \frac{2 * Precision * Recall}{Precision + Recall} \end{aligned}$$

1. True Positive (TP): The number of instances correctly predicted as positive.
2. True Negative (TN): The number of instances correctly predicted as negative.
3. False Positive (FP): The number of instances incorrectly predicted as positive (Type I error).
4. False Negative (FN): The number of instances incorrectly predicted as negative (Type II error).

Figure 9,10,11,12 and 13 show the confusion matrix of each machine learning model used that are XGBoost, LightGBM, CatBoost, L1 Regularization and Fast Tree. The confusion matrix is crucial to provide a thorough and precise analysis of a model's performance. It allows a more thorough comprehension of the model's performance with regard to producing accurate and inaccurate predictions across various attributes.

Five distinct machine learning models, including XGBoost, LightGBM, CatBoost, L1 Regularization and Fast Tree, were deployed to forecast the various outcomes of the student's adaptability level in this study. The objective was to get a more reliable predictive model by comparing the performance of these models. This study used 80% of the data as a training set and the remaining 20% as a test set. 3by3 confusion matrices are generated to conclude the large dataset collected from Kaggle. Generated confusion matrix by the applied models based on the dataset is presented in Figure 9,10,11,12 and 13.

To assess the model's performance, four sorts of evaluation pointers are employed: Precision, Recall, Accuracy, and F1-Score. Table 2,3,4 and 5 shows the accuracy, precision, recall, and F1-score produced by the various models. It is observed that the Fast Tree model has the top accuracy rate of 0.91 of the five machine learning models employed for prediction, while the L1 Regularization has the lowest accuracy rate of 0.69.

Table 7: Shows the ranking of the model's accuracies

Model	Accuracy	Rank
FastTree	0.912863	1
XGBoost	0.746888	2
LightGBM	0.716805	3
CatBoost	0.703320	4
L1 Regularization	0.688797	5

For the majority of real-world circumstances, when an uneven class distribution is the norm, the F1-score is a more appropriate metric for assessing the model. The models perform better when the F1 values are higher. From table 2,3,4 and 5, it is found out that Fast Tree has the highest F1-scores of 0.75, 0.94 and 0.92 for all target attributes of High Adaptability, Low Adaptability and Moderate Adaptability. LightGBM has the lowest F1-score for High Adaptability of 0.40. CatBoost has the lowest F1-score for Low Adaptability of 0.64 and L1 Regularization has the lowest F1-score for Moderate Adaptability of 0.47. This indicates that the Fast Tree model has better performance than other models.

Table 7 and Figure 14 display the summary of machine learning models performance metrics. The mean accuracy of 0.75 indicates the overall average performance level across the models. A higher mean accuracy suggests that, on average, the models perform well in predicting outcomes. The minimum and maximum accuracy scores of 0.69 and 0.91 respectively highlight the variability in performance among the models, with the highest-scoring model achieving significantly better results than the lowest-scoring one. The mode accuracy of 0.69 indicates that this specific accuracy value is the most common among the models, suggesting a potential area where model performance might converge. The standard deviation of approximately 0.0842 suggests moderate variability in accuracy scores around the mean, indicating that while the models, on average,

perform well, there is some dispersion in their performance levels. Lastly, the 95% confidence interval of (0.6467, 0.8559) provides a range within which the true mean accuracy of all models is estimated to lie with 95% confidence. This interval helps assess the precision of accuracy estimation and provides insights into the reliability of the observed mean accuracy. Overall, these statistics aid in understanding the distribution, variability, and reliability of model performance, enabling better-informed decisions in model selection and deployment.

Table 8 displays a T-test for comparing the means of accuracy of each Machine Learning Models. The t-test results reveal significant differences in accuracy between all pairs of machine learning models. Each comparison yielded a very low p-value (approximately 0.0000), indicating strong evidence against the null hypothesis of equal means. Starting with the comparison between L1 Regularization and other models, notably, the t-statistics are remarkably large, indicating substantial differences in mean accuracies. For instance, the t-statistic between L1 Regularization and FastTree is $-9.91e+16$, suggesting a vast disparity in performance. Moving to comparisons between other models, such as XGBoost, CatBoost, and LightGBM, the t-statistics remain sizable, with values ranging in the order of 10^{16} . This trend persists across all comparisons, indicating that each model offers unique predictive capabilities. Interestingly, the t-statistic between XGBoost and FastTree yields negative infinity, suggesting an exceptionally large difference in mean accuracies. These findings underscore the importance of model selection, as the choice can significantly impact predictive performance.

From the above analysis of the result, it is perceived that the Fast Tree model outperformed the other models in terms of performance metrics. It is also evident from the accuracy claims made that the Fast Tree model produces high-quality results. Therefore, in online education, the Fast Tree model may be a suitable option for predicting students' adaptability.

9.0 Conclusion

This study explores the ever changing world of online learning using a powerful classification-based model to examine student adaptability. Drawing on data collected from the Kaggle platform, five different machine learning models are used. Fast Tree with accuracy of 0.91, XGBoost with accuracy of 0.75, lightGBM with accuracy of 0.72, L1 Regularization with accuracy of 0.69 and CatBoost with accuracy of 0.70 to uncover the complex patterns and trends that shape student success in an online environment. These results emphasize the importance of generalizability over training performance when assessing model effectiveness.

Picking five distinct models as a model selection was an ingenious manner to compare their performance and understand student adaptability. Even though Fast Tree was extremely accurate in their training, the generalization is also an important factor to consider as shown by XGBoost and LightGBM superior test results. This is consistent with the content of Probability and Statistics about the risks of overfitting and testing models on unseen data. Multiple metrics beyond just accuracy, such as precision, recall and F1 score would offer a more comprehensive view of each model's strong points and the weaknesses. This is linked to metrics such as true positive rates and false positive rates from Probability and Statistics, which provide a more detailed evaluation of the model's performance. The approach of analysis through descriptive and inferential statistics, probability distributions, graphical representation is a reliable way to get more insights than just model prediction. It provides an opportunity to look at features such as central tendencies, variance and relationships between variables which directly relate to the contents of the Probability and Statistics course.

According to the study's results based on complex relationships among factors as age, education level, location, load-shedding and financial condition influencing student adaptability underlie advanced perception of data analysis established for consideration. This is a manifest of probability distributions and the possibility to represent multivariate relationships within datasets. This classification provides a way to allow specific interventions, content and support resources that will be effective in different groups allowing the educators or policymakers to best serve all students working within this dynamic digital environment.

This study provides valuable knowledge that can be beneficial for both teachers and policymakers in order to appreciate the opportunities arising from using online learning. Taking

into account that multiple dynamics affect the ability of individuals to adapt, an awareness in this regard can facilitate the construction of inclusive learning environments; promote diversity and eventually provide learners with tools needed for achieving success in a digitalized age. This study is quite informative for educational practitioners as well as those developers and policymakers who need this information in order to benefit effectively from the advantages of online learning. A complete knowledge of all the factors that influence students' adaptive skills gives an opportunity to create favorable learning conditions, build diverse classrooms and finally provide the necessary tools for success in today's digital world.

10.0 References

M. Hasan Suzan, N. A. Samrin, A. A. Biswas and A. Pramanik, "Students' Adaptability Level Prediction in Online Education using Machine Learning Approaches," 2021 12th International Conference on Computing Communication and Networking Technologies (ICCCNT), Kharagpur, India, 2021, pp. 1-7, <https://doi.org/10.1109/ICCCNT51525.2021.9579741>

Gligorea, Ilie, Marius Cioca, Romana Oancea, Andra-Teodora Gorski, Hortensia Gorski, and Paul Tudorache. 2023. "Adaptive Learning Using Artificial Intelligence in e-Learning: A Literature Review" *Education Sciences* 13, no. 12: 1216 <https://doi.org/10.3390/educsci13121216>

Besser, A., Flett, G. L., & Zeigler-Hill, V. (2022). Adaptability to a sudden transition to online learning during the COVID-19 pandemic: Understanding the challenges for students. *Scholarship of Teaching and Learning in Psychology*, 8(2), 85–105. <https://doi.org/10.1037/stl0000198>

He, Wu; Xu, Guandong; and Kruck, S. E. (2014) "Online IS Education for the 21st Century," *Journal of Information Systems Education*: Vol. 25 : Iss. 2 , 101-106. Available at: <https://aisel.aisnet.org/jise/vol25/iss2/1>

Mayadas, A. F., Bourne, J., & Bacsich, P. (2009). Online Education Today. *Science*, 323(5910), 85–89. <https://doi.org/10.1126/science.1168874>

M. Hasan Suzan, N. A. Samrin, A. A. Biswas and A. Pramanik, "Students' Adaptability Level Prediction in Online Education using Machine Learning Approaches," 2021 12th International Conference on Computing Communication and Networking Technologies (ICCCNT), Kharagpur, India, 2021, pp. 1-7, doi: 10.1109/ICCCNT51525.2021.9579741.

M. Tan and P. Shao, "Prediction of student dropout in e-learning program through the use of machine learning method", *International Journal of Emerging Technologies in Learning*, vol. 10, no. 1, 2015.

Singh, R. N., & Hurley, D. (2017). The Effectiveness of Teaching and Learning Process in Online Education as Perceived by University Faculty and Instructional Technology Professionals. *Journal of Teaching and Learning with Technology*, 6(1), 65–75. <https://doi.org/10.14434/jotlt.v6.n1.19528>

Students Adaptability Level in Online Education. (n.d.). [www.kaggle.com](https://www.kaggle.com/datasets/mdmahmudulhasansuzan/students-adaptability-level-in-online-education). <https://www.kaggle.com/datasets/mdmahmudulhasansuzan/students-adaptability-level-in-online-education>

V. L. Uskov, J. P. Bakken, A. Byerly and A. Shah, "Machine learning-based predictive analytics of student academic performance in stem education", *2019 IEEE Global Engineering Education Conference (EDUCON)*, pp. 1370-1376, 2019.

Xiaoyan Xie , Fuzong Lin & Tao Zhang (2001) Comparison between on- and off-campus behaviour and adaptability in online learning: A case from China, *Behaviour & Information Technology*, 20:4, 281-291, DOI: 10.1080/01449290110083594

Yavuz, M. (2023, April 22). *Popular Machine Learning Algorithms: CART, Random Forest, GBM, XGBoost, LightGBM, and CatBoost*. Medium. <https://medium.com/@arcyalcin/popular-machine-learning-algorithms-cart-random-forest-gbm-xgboost-lightgbm-and-catboost-2c22018cdef>